

Curriculum Vitae

Simone Manti

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EDUCATION

Technical University of Denmark (DTU)	Copenhagen, Denmark
Ph.D. in Physics , Supervisor: Kristian Sommer Thygesen	May 2018–Sep. 2021
– Thesis: “Effects of lattice imperfections on the optical and electronic properties of two-dimensional materials”	
University of Roma Tor Vergata	Rome, Italy
M.S. in Physics 110 e lode/110, Supervisor: Gianluca Stefanucci	Apr. 2016–Dec. 2017
– Thesis: “Real time photoionization of atoms: Successes and limitation of the Non Equilibrium Greens Functions method”	
University of Roma Tor Vergata	Rome, Italy
Passage course from Material Science to Physics	Ott. 2015–Apr. 2016
University of Roma Tor Vergata	Rome, Italy
B.S. in Material Science 104/110, Supervisors: Anna Sgarlata and Massimo Fanfoni	Ott. 2012–Ott. 2015
– Thesis: “Study and characterization of the formation of crystal defects on surfaces of Germanium with different orientation”	

WORK & RESEARCH EXPERIENCE

INFN-LNF	Frascati, Italy
Assegnista di Ricerca, Supervisor: Catalina Oana Curceanu	May 2022–Now
Technical University of Denmark (DTU)	Copenhagen, Denmark
Research Assistant, Supervisor: Kristian Sommer Thygesen	Sep. 2021–Apr. 2022

SCIENTIFIC INTERESTS

ED-XRF; WD-XRF; MCDF; XES; Kaonic Atoms; Atomic Cascade Calculation; Ab-initio codes; High-Throughput Studies; Databases; Machine Learning;

TEACHING EXPERIENCE

Electronic Structure Methods in Material Physics (10302)	Spring 2019
<i>Teaching Assistant at Technical University of Denmark</i>	
Condensed Matter Physics and Nanoscale Materials Physics (10303)	Fall 2018
<i>Teaching Assistant at Technical University of Denmark</i>	

VISITING SCIENTIST & PRACTICES

Wigner Research Centre for Physics

Visit, under the supervision of Lajos Diosi

Budapest, Hungary

January 2024

Sorbonne Université

Visit, under the supervision of Paul Indelicato

Paris, France

December 2023

MLHEP 2023 Summer School

School on Machine Learning in High Energy Physics

Erice, Italy

April 2023

Center for Physical Sciences and Technology (FTMC)

External stay during the PhD, under the supervision of Audrius Alkauskas

Vilnius, Lithuania

Sep. 2019–Dec. 2019

CAMD Summer School

School on Electronic Structure Theory and Materials Design

Helsingør, Denmark

August 2018

RESEARCH ACTIVITIES

MITIQO (2022-Now): I am the responsible for the implementation of the MITIQO project financed by Regione Lazio, Progetti di Ricerca. The project aimed to study metals in edible liquids such as wine and olive oil to determine their quality and geographical origin using X-ray Fluorescence (XRF). For quality assessment, XRF was performed in Wavelength-Dispersive (WD) mode, as determining the quality requires finely resolving the emission lines of metals. The main challenge of WD-XRF is the low emission rate for samples with trace amounts of metal. I contributed to improving the limit of detection of the technique by optimizing the setup and developing algorithms for background reduction to enhance the signal-to-background ratio. For determining the geographical origin of edibles, I used XRF in Energy-Dispersive (ED) mode to measure the concentration of metals in a wide energy range. I have built a dataset for metal concentrations for a set of wines from the Lazio province. I proposed to infer the geographical origin by applying multivariate analysis techniques, such as Principal Component Analysis, to reduce the dimensionality of the dataset. This allowed me to determine clusters in the dataset and then fit machine learning models for classifying wine depending on the geographical origin.

SIDDHARTA2 (2022-Now): I am a member of the SIDDHARTA2 collaboration at LNF, where my primary activity was simulating the X-ray emission spectra of kaonic atoms. I employed Multi-Configuration Dirac-Fock (MCDF) code to simulate transitions for various purposes. I used them to develop a cascade code that accounts for the relevant effects of the kaonic atom interactions during the cascade process, allowing for accurate simulation of the X-ray transition yields. Additionally, I used this code to theoretically investigate and place bounds on the kaon mass, in combination with results from experiments. I also took part in the experimental phase, where I applied my knowledge of machine learning to perform unsupervised anomaly detection for X-ray spectra using autoencoders.

VIP (2022-Now): For the VIP-related experiments at the National Laboratory of Gran Sasso, I performed calculations from first principles, also using the MCDF code, to calculate the transition energy for electrons violating the Pauli Exclusion Principle. Additionally, I was involved in developing models to describe spontaneous collapse. Specifically, I included the collapse model effect on the atom, calculating atomic properties using first-principles code.

PHD (2018-2021): During my PhD, my focus was on predicting the properties of two-dimensional materials using ab-initio computational methods. I was also involved in creating databases of 2D materials obtained through high-throughput techniques by combining existing materials to guide the discovery of new ones. All the results were organized into databases, where I applied various analyses to study their properties. For example, we introduced defects in materials and calculated the emission lines of these systems due to their potential as single-photon

emitters or qubits. Additionally, I applied machine learning to predict the dynamic stability of the proposed materials.

RESPONSABILITIES

1. For the **MITIQO** project, I was responsible for maintaining the X-ray fluorescence setup and performing the measurements. During the project, I extensively broadened the scope by implementing optimization methods to enhance the system's sensitivity towards the detection of trace metals.
2. **XRF**: Conducted detailed analysis using X-ray fluorescence to investigate material composition, identify contaminants, and quantify trace elements. Developed protocols to improve the accuracy and reproducibility of XRF measurements.
3. **Atomic Spectra**: Analyzed atomic spectra to identify and quantify elements within samples. Applied advanced spectroscopic techniques to study energy levels and transitions, contributing to fundamental research in atomic and nuclear physics.
4. **Machine Learning**: Utilized and developed machine learning algorithms to analyze complex datasets and identify patterns. Implemented predictive models to enhance research outcomes and optimize experimental procedures.
5. Active member of the educational team of the LNF-INFN: activities for students at Liceo B. Touschek and Winter Stages for "Incontri di Fisica" for high school science teachers.

CONFERENCES, WORKSHOPS & SEMINARS

A Modern Odyssey: Quantum Gravity meets Quantum Collapse at Atomic and Nuclear physics energy scales in the Cosmic Silence, ECT*

Trento, Italy

June 2024

- Invited talk: "Cancellation Effects in the Spontaneous Collapse Rate at Low Energy Limit"

ALPACA: modern ALgorithms in machine learning and data analysis from medical Physics to research with ACcelerAtors and in underground laboratories, ECT*

Trento, Italy

November 2023

- Invited talk: "ML in SIDDHARTA-2: the monitoring challenge"

**Quantum Information and Probability:
from Foundations to Engineering (QIP23)**

Linnaeus University, Växjö, Sweden

June 2023

- Talk: "Underground tests of Quantum Mechanics: Collapse models and Pauli Exclusion Principle"

EXOTICO: EXOTic atoms meet nuclear COLLisions for a new frontier precision era in low-energy strangeness nuclear physics, ECT*

Trento, Italy

October 2022

- Invited talk: "Cascade models for atomic transitions in kaonic atoms"

Nuclear and Atomic Transitions as Laboratories for High Precision Tests of Quantum Gravity Inspired Models, ECT*

Trento, Italy

September 2022

- Invited talk: "Cascade models for atomic transitions"

Seminar at Center for Physical Sciences and Technology (FTMC)

Vilnius, Lithuania

Nov. 2019

- Seminar: “High-throughput material discovery: The Computational 2D Materials Database”

Deutsche Physikalische Gesellschaft (DPG) Spring Meeting

Regensburg, Germany
Apr. 2019

- Talk: “Controlling band alignment at heterointerfaces using atomically thin Janus structures with built-in dipoles”

Progress in Nonequilibrium Green’s Functions

Frascati, Italy
August 2018

- Poster: “Nonradiative recombination processes in semiconductors for opto-electronics”

LANGUAGES

- **Italian** Native proficiency
- **English** Professional working proficiency
- **Danish** Elementary proficiency

LIST OF PUBLICATIONS

- [1] F. Artibani, F. Clozza, M. Bazzi, C. Capoccia, A. Clozza, L. De Paolis, K. Dulski, C. Guaraldo, M. Iliescu, Khreptak, **S. Manti**, *et al.*, “The odyssey of kaonic atoms studies at the dafne collider: From dear to siddharta-2.”, *Acta Physica Polonica B*, vol. 55, no. 5, 2024.
- [2] D. Bosnar, L. Abbene, C. Amsler, F. Artibani, M. Bazzi, M. Bragadireanu, A. Buttacavoli, M. Cargnelli, M. Carminati, A. Clozza, F. Clozza, G. Deda, L. D. Paolis, R. D. Grande, K. Dulski, L. Fabbietti, C. Fiorini, I. Frii, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, M. Makek, **S. Manti**, J. Marton, P. Moskal, F. Napolitano, S. Niedwiecki, H. Ohnishi, K. Piscicchia, A. Scordo, F. Sgaramella, D. Sirghi, F. Sirghi, M. Skurzok, M. Silarski, A. Spallone, K. Toho, M. Tüchler, O. V. Doce, J. Zmeskal, and C. Curceanu, *Kaonic lead feasibility measurement at dafne to solve the charged kaon mass discrepancy*, 2024. arXiv: 2405.12942.
- [3] De Paolis, L., Abbene, L., Bazzi, M., Bettelli, M., Borghi, G., Bosnar, D., Bragadireanu, M., Buttacavoli, A., Cargnelli, M., Carminati, M., Clozza, A., Deda, G., Del Grande, R., Dulski, K., Fiorini, C., Frii, I., Guaraldo, C., Iliescu, M., Iwasaki, M., Khreptak, A., **S. Manti**, Marton, J., Miliucci, M., Moskal, P., Napolitano, F., Niedzwiecki, S., Obertova, J., Ohnishi, O., Piscicchia, K., Principato, F., Sada, Y., Scordo, A., Sgaramella, F., Silarski, M., Sirghi, D.L., Sirghi, F., Skurzok, M., Wycech, S., Spallone, A., Toho, K., Tüchler, M., Yoshida, C., Zappettini, A., Zmeskal, J., and Curceanu, C., “The measurement of the e^2 nuclear resonance effects in kaonic atoms at dane: The kameo proposal”, *EPJ Web Conf.*, vol. 291, p. 05 003, 2024.
- [4] **S. Manti**, F. Napolitano, A. Clozza, C. Curceanu, G. Moskal, K. Piscicchia, D. Sirghi, and A. Scordo, “Enhancing performances of the voxes bragg spectrometer for xes investigations”, *Condensed Matter*, vol. 9, no. 1, 2024, ISSN: 2410-3896.
- [5] H. Moustafa, J. Kangsabanik, F. Bertoldo, **S. Manti**, K. S. Thygesen, K. W. Jacobsen, and T. Olsen, “Selenium and the role of defects for photovoltaic applications”, *Phys. Rev. Mater.*, vol. 8, p. 015 402, 1 Jan. 2024.
- [6] K. Piscicchia, A. Clozza, D. L. Sirghi, M. Bazzi, N. Bortolotti, M. Bragadireanu, M. Cargnelli, L. De Paolis, R. Del Grande, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, J. Marton, M. Miliucci, F. Napolitano, A. Porcelli, A. Scordo, F. Sgaramella, F. Sirghi, S. Tomassini, O. V. Doce, J. Zmeskal, and C. Curceanu, “Optimization of a beige detector setup for testing quantum foundations in the underground Ings laboratory”, *Condensed Matter*, vol. 9, no. 2, 2024, ISSN: 2410-3896.

- [7] A. Porcelli, M. Bazzi, N. Bortolotti, M. Bragadireanu, M. Cargnelli, A. Clozza, L. De Paolis, R. Del Grande, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, J. Marton, M. Miliucci, F. Napolitano, K. Piscicchia, A. Scordo, F. Sgaramella, D. L. Sirghi, F. Sirghi, O. V. Doce, J. Zmeskal, and C. Curceanu, “Vip-2 with modulated current: Pathfinder for enhanced pauli exclusion principle violation studies”, *The European Physical Journal C*, vol. 84, no. 3, p. 214, Mar. 2024, ISSN: 1434-6052.
- [8] F. Sgaramella *et al.*, “First measurement of kaonic helium-4 M-series transitions”, *J. Phys. G*, vol. 51, no. 5, p. 055 103, 2024. arXiv: 2310.20584 [nucl-ex].
- [9] F. Sgaramella *et al.*, “Kaonic Helium-4 *L*-series Yield Measurement at 2.25 g/l Density by SIDDHARTA-2 at DAΦNE”, *Acta Phys. Polon. Supp.*, vol. 17, no. 1, 1–A8, 2024.
- [10] F. Sgaramella, F. Clozza, L. Abbene, F. Artibani, M. Bazzi, G. Borghi, M. Bragadireanu, A. Buttacavoli, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, R. Del Grande, L. De Paolis, K. Dulski, C. Fiorini, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Ohnishi, K. Piscicchia, F. Principato, A. Scordo, M. Silarski, D. Sirghi, F. Sirghi, M. Skurzok, A. Spallone, K. Toho, M. Tüchler, J. Zmeskal, and C. Curceanu, “Characterization of the siddharta-2 setup via the kaonic helium measurement”, *Condensed Matter*, vol. 9, no. 1, 2024, ISSN: 2410-3896.
- [11] Sirghi, F., Abbene, L., Bazzi, M., Bosnar, D., Bragadireanu, M., Buttacavoli, A., Cargnelli, M., Carminati, M., Clozza, A., Deda, G., Del Grande, R., Dulski, K., De Paolis, L., Fabbietti, L., Fiorini, C., Frii, I., Guaraldo, C., Iliescu, M., Iwasaki, M., Khreptak, A., **S. Manti**, Marton, J., Miliucci, M., Moskal, P., Napolitano, F., Sgaramella, F., Niedwiecki, S., Ohnishi, H., Piscicchia, K., Sada, Y., Scordo, A., Silarski, M., Sirghi, D., Skurzok, M., Spallone, A., Toho, K., Tüchler, M., Yoshida, C., Zmeska, J., and Curceanu, C., “Kaonic atoms with siddharta-2 at the dafne collider”, *EPJ Web Conf.*, vol. 291, p. 01 008, 2024.
- [12] L. Abbene, M. Bettelli, A. Buttacavoli, F. Principato, A. Zappettini, C. Amsler, M. Bazzi, D. Bosnar, M. Bragadireanu, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, L. De Paolis, R. Del Grande, L. Fabbietti, C. Fiorini, I. Frii, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Ohnishi, K. Piscicchia, Y. Sada, F. Sgaramella, H. Shi, M. Silarski, D. L. Sirghi, F. Sirghi, M. Skurzok, A. Spallone, K. Toho, M. Tüchler, O. V. Doce, C. Yoshida, J. Zmeskal, A. Scordo, and C. Curceanu, “New opportunities for kaonic atoms measurements from cdznte detectors”, *The European Physical Journal Special Topics*, vol. 232, no. 10, pp. 1487–1492, Aug. 2023, ISSN: 1951-6401.
- [13] L. Abbene, A. Buttacavoli, F. Principato, G. Gerardi, M. Bettelli, A. Zappettini, M. Bazzi, M. Bragadireanu, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, R. Del Grande, L. De Paolis, L. Fabbietti, C. Fiorini, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Ohnishi, K. Piscicchia, Y. Sada, F. Sgaramella, H. Shi, M. Silarski, D. L. Sirghi, F. Sirghi, M. Skurzok, A. Spallone, K. Toho, M. Tüchler, O. V. Doce, C. Yoshida, J. Zmeskal, A. Scordo, and C. Curceanu, “Potentialities of cdznte quasi-hemispherical detectors for hard x-ray spectroscopy of kaonic atoms at the dane collider”, *Sensors*, vol. 23, no. 17, 2023, ISSN: 1424-8220.
- [14] S. Ali, F. A. Nilsson, **S. Manti**, F. Bertoldo, J. J. Mortensen, and K. S. Thygesen, “High-throughput search for triplet point defects with narrow emission lines in 2d materials”, *ACS Nano*, vol. 17, no. 21, pp. 21 105–21 115, 2023, PMID: 37889165.
- [15] C. Curceanu, L. Abbene, C. Amsler, M. Bazzi, M. Bettelli, G. Borghi, D. Bosnar, M. Bragadireanu, A. Buttacavoli, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, R. Del Grande, L. De Paolis, K. Dulski, C. Fiorini, I. Frii, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Onishi, K. Piscicchia, F. Principato, Y. Sada, A. Scordo, F. Sgaramella, H. Shi, M. Silarski, D. L. Sirghi, F. Sirghi, M. Skurzok, A. Spallone, K. Toho, M. Tüchler, O. V. Doce, C. Yoshida, A. Zappettini, and J. Zmeskal, “Kaonic atoms at the dane collider: A strangeness adventure”, *Frontiers in Physics*, vol. 11, 2023, ISSN: 2296-424X.

- [16] C. Curceanu, L. De Paolis, M. Bazzi, D. Bosnar, M. Bragadireanu, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, K. Dulski, C. Fiorini, I. Frieri, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, P. Levi Sandri, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Onishi, K. Piscicchia, Y. Sada, A. Scordo, F. Sgaramella, M. Silarski, D. L. Sirghi, F. C. Sirghi, M. Skurzok, A. Spallone, K. Toho, M. Tüchler, C. Yoshida, and J. Zmeskal, "Towards the first kaonic deuterium measurement with the SIDDHARTA-2 experiment", 39, p. 39, Jan. 2023.
- [17] L. De Paolis, M. Bazzi, D. Bosnar, M. Bragadireanu, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, R. Del Grande, K. Dulski, C. Fiorini, I. Frieri, C. Guaraldo, M. Iliescu, M. Iwasaki, A. Khreptak, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, O. Onishi, K. Piscicchia, Y. Sada, A. Scordo, F. Sgaramella, H. Shi, M. Silarski, D. L. Sirghi, F. Sirghi, M. Skurzok, S. Wycech, A. Spallone, K. Toho, M. Tüchler, C. Yoshida, J. Zmeskal, and C. Curceanu, "The cameo proposal : Investigation of the e2 nuclear resonance effects in kaonic atoms", eng, *Il Nuovo Cimento C*, vol. 46, no. 3, 2023, ISSN: 2037-4909.
- [18] De Paolis, L., Bosnar, D., Bragadireanu, M., Cargnelli, M., Carminati, M., Clozza, A., Deda, G., Del Grande, R., Dulski, K., Fiorini, C., Frieri, I., Guaraldo, C., Iliescu, M., Iwasaki, M., Khreptak, A., **S. Manti**, Marton, J., Miliucci, M., Moskal, P., Napolitano, F., Niedwiecki, S., Onishi, O., Piscicchia, K., Sada, Y., Scordo, A., Sgaramella, F., Silarski, M., Sirghi, D.L., Sirghi, F., Skurzok, M., Wycech, S., Spallone, A., Toho, K., Tüchler, M., Yoshida, C., Zmeskal, J., and Curceanu, C., "Investigating the e2 nuclear resonance effects in kaonic atoms: The cameo proposal", *EPJ Web of Conf.*, vol. 290, p. 06 003, 2023.
- [19] **S. Manti**, M. Miliucci, A. Scordo, R. Bedogni, A. Clozza, M. Iliescu, G. Moskal, K. Piscicchia, A. Porcelli, D. Sirghi, F. Sirghi, and C. Curceanu, *Detecting iron oxidation states in liquids with the voxes bragg spectrometer*, 2023. arXiv: 2310.06062.
- [20] **S. Manti**, M. K. Svendsen, N. R. Knøsgaard, P. M. Lyngby, and K. S. Thygesen, "Exploring and machine learning structural instabilities in 2d materials", *npj Computational Materials*, vol. 9, no. 1, p. 33, Mar. 2023, ISSN: 2057-3960.
- [21] F. Napolitano, A. Addazi, A. Bassi, M. Bazzi, M. Bragadireanu, M. Cargnelli, A. Clozza, L. De Paolis, R. Del Grande, M. Derakhshani, S. Donadi, C. Fiorini, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, A. Marcianò, J. Marton, M. Miliucci, E. Milotti, K. Piscicchia, A. Porcelli, A. Scordo, F. Sgaramella, D. L. Sirghi, F. Sirghi, O. V. Doce, J. Zmeskal, and C. Curceanu, "Underground tests of quantum mechanics by the vip collaboration at gran sasso", *Symmetry*, vol. 15, no. 2, 2023, ISSN: 2073-8994.
- [22] F. Napolitano, M. Bazzi, M. Bragadireanu, M. Cargnelli, A. Clozza, L. D. Paolis, R. D. Grande, C. Fiorini, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, J. Marton, M. Miliucci, K. Piscicchia, A. Porcelli, A. Scordo, F. Sgaramella, D. L. Sirghi, F. Sirghi, O. V. Doce, J. Zmeskal, and C. Curceanu, "Novel machine learning and differentiable programming techniques applied to the vip-2 underground experiment", *Measurement Science and Technology*, vol. 35, no. 2, p. 025 501, Nov. 2023.
- [23] L. D. Paolis, M. Bazzi, D. Bosnar, M. Bragadireanu, M. Cargnelli, M. Carminati, A. Clozza, G. Deda, R. D. Grande, K. Dulski, C. Fiorini, I. Frieri, C. Guaraldo, M. Iliescu, M. Iwasaki, P. King, A. Khreptak, P. L. Sandri, **S. Manti**, J. Marton, M. Miliucci, P. Moskal, F. Napolitano, S. Niedwiecki, H. Onishi, K. Piscicchia, Y. Sada, A. Scordo, F. Sgaramella, H. Shi, M. Silarski, D. L. Sirghi, F. Sirghi, M. Skurzok, S. Wycech, A. Spallone, K. Toho, M. Tüchler, O. V. Doce, C. Yoshida, J. Zmeskal, and C. Curceanu, "Investigating the e2 nuclear resonance effect in kaonic atoms", *Journal of Physics: Conference Series*, vol. 2446, no. 1, p. 012 038, Feb. 2023.
- [24] K. Piscicchia, S. Donadi, **S. Manti**, A. Bassi, M. Derakhshani, L. Diosi, and C. Curceanu, "Cancellation effects as a fingerprint of quantum collapse models at atomic scale", 2023. arXiv: 2301.09920.
- [25] K. Piscicchia, A. Marcianò, A. Addazi, D. L. Sirghi, M. Bazzi, N. Bortolotti, M. Bragadireanu, M. Cargnelli, A. Clozza, L. De Paolis, R. Del Grande, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, J. Marton, M. Miliucci, F. Napolitano, F. Nola, A. Porcelli, A. Scordo, F. Sgaramella, F. Sirghi, O. V. Doce, J. Zmeskal, and

- C. Curceanu, "First experimental survey of a whole class of non-commutative quantum gravity models in the vip-2 lead underground experiment", *Universe*, vol. 9, no. 7, 2023, ISSN: 2218-1997.
- [26] K. Piscicchia, A. Porcelli, A. Bassi, M. Bazzi, M. Bragadireanu, M. Cargnelli, A. Clozza, L. De Paolis, R. Del Grande, M. Derakhshani, D. Lajos, S. Donadi, C. Guaraldo, M. Iliescu, M. Laubenstein, **S. Manti**, J. Marton, M. Miliucci, F. Napolitano, A. Scordo, F. Sgaramella, D. L. Sirghi, F. Sirghi, O. Vazquez Doce, J. Zmeskal, and C. Curceanu, "A novel approach to parameter determination of the continuous spontaneous localization collapse model", *Entropy*, vol. 25, no. 2, 2023, ISSN: 1099-4300.
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Frascati, 15/06/2024

Simone Manti

A handwritten signature in dark ink, reading "Simone Manti". The script is fluid and cursive, with the first name and last name clearly distinguishable.